

2. Suppose we are developing a music playlist application. The application has a playlist that contains multiple music tracks. We want to implement the functionality to iterate over the tracks in the playlist, allowing users to play each track one by one, skip tracks, and get the current track information. The playlist has three main components: the `MusicPlaylist` class which manages the collection of tracks, the `Track` class representing each individual music track with properties like title, artist, and duration, and the `MusicPlaylistIterator` class which is the iterator for traversing the playlist. The client program is named `PlaylistTester`.

(Hint : The `MusicPlaylistIterator` should provide methods such as `hasNext()` to check if there is a next track, `next()` to move to the next track and return it, and `reset()` to start the iteration from the beginning. The `MusicPlaylist` class should have methods to add tracks and get the iterator.)

MusicPlaylist

```
package experiment18_2;

import java.util.ArrayList;
import java.util.List;

class MusicPlaylist {
```

```

    private List<Track> trackList;

    public MusicPlaylist(){
        this.trackList = new ArrayList<Track>();
    }

    public void addTrack(Track track){
        trackList.add(track);
    }

    public MusicPlaylistIterator getIterator(){
        return new MusicPlaylistIterator(trackList);
    }
}

```

MusicPlaylistIterator

```

package experiment18_2;

import java.util.ArrayList;
import java.util.List;

class MusicPlaylistIterator {
    private List<Track> trackList;
    private int position;

    public MusicPlaylistIterator(List<Track> trackList){
        this.trackList = trackList;
    }
}

```

```

        position = 0;
    }

    public boolean hasNext(){
        if(position < trackList.size()){
            return true;
        }else{
            return false;
        }
    }

    public Track next(){
        return trackList.get(position++);
    }

    public void reset(){
        position = 0;
    }
}

```

PlaylistTester

```

package experiment18_2;

public class PlaylistTester {
    public static void main(String[] args) {
        MusicPlaylist musicPlaylist = new MusicPlaylist();
    }
}

```

```

Track track1 = new Track("蜃楼", "周深", "3'46'");
Track track2 = new Track("嗨", "周深", "4'00'");
Track track3 = new Track("虚构", "周深", "5'56'");
musicPlaylist.addTrack(track1);
musicPlaylist.addTrack(track2);
musicPlaylist.addTrack(track3);

MusicPlaylistIterator iterator =
musicPlaylist.getIterator();

while (iterator.hasNext()){
    Track currentTrack = iterator.next();
    System.out.println("Playing: " + currentTrack);
}

iterator.reset();

System.out.println("Reset playlist.");

System.out.println("Now playing: " + iterator.next());
}
}

```

Track

```

package experiment18_2;

class Track {

```

```
private String title;

private String artist;

private String duration;

public Track(String title, String artist, String duration){

    this.title = title;

    this.artist = artist;

    this.duration = duration;

}

public String getTitle(){

    return title;

}


public String getArtist() {

    return artist;

}


public String getDuration() {

    return duration;

}

@Override

public String toString() {

    return "Track: " + title + ", Artist: " + artist + ",
```

```
Duration: " + duration;  
  
    }  
  
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ IDEA 2021.2.  
Playing: Track: 蜃楼, Artist: 周深, Duration: 3'46''  
Playing: Track: 嗨, Artist: 周深, Duration: 4'00''  
Playing: Track: 虚构, Artist: 周深, Duration: 5'56''  
Reset playlist.  
Now playing: Track: 蜃楼, Artist: 周深, Duration: 3'46''  
  
Process finished with exit code 0
```